

PROJECT REPORT

FORESIGHT

Demand & Inventory Intelligence

Client: NorthBay Living

Project objective	Forecast weekly SKU demand and convert forecast signals into inventory actions.
Dataset	Clean synthetic retail transaction dataset with supplied SKU and inventory snapshot tables.
Scope	5,000 SKUs; weekly demand forecasting; 8-week outlook; inventory decision support.
Prepared for	Internship project submission

September 2026

1. Project Overview

Project FORESIGHT is a demand forecasting and inventory decision-support solution built for NorthBay Living. The system aggregates retail transactions into weekly SKU demand, compares a machine-learning forecast against a seasonal-naive baseline, and translates an eight-week forecast into practical inventory actions.

Business Problem

Retail teams need a clear and repeatable way to identify where demand may exceed available stock and where inventory may be excessive. FORESIGHT supports this decision by pairing SKU-level demand forecasts with an inventory snapshot and transparent risk logic.

Objectives

- Forecast SKU-level weekly demand for the next eight weeks.
- Benchmark the model against a seasonal-naive baseline using WAPE.
- Identify stockout and overstock risk using supplied inventory snapshot fields.
- Recommend REORDER NOW, MARKDOWN / CLEAR, WATCH / VOLATILE, or HEALTHY actions.
- Present results in a Streamlit dashboard for business users.

2. Dataset and Data Audit

The project uses the clean version of the supplied synthetic retail dataset. Processing was designed to be memory-efficient: only needed columns were read and raw transactions were aggregated early to SKU-week demand.

Audit item	Actual result
Transaction rows	9,972,038
Transaction fields	11
Date range	2022-01-01 to 2025-12-31
Unique SKUs	5,000
Total units sold	18,750,830
Total recorded revenue	10,917,737,293 monetary units
Promotion ID missing	7,881,687 rows (a blank promo ID indicates no linked promotion)

Data Mapping

FORESIGHT field	Source dataset field	Availability
SKU	sales_transactions.sku_id	Available
Demand	sales_transactions.quantity	Available
Revenue	sales_transactions.total_value	Available
Category	sku_master.category	Available
Promotion	promo_id and discount_pct	Available
Inventory	inventory_snapshot.stock_on_hand, reorder_point, safety_stock	Available - snapshot only
On order	Not supplied	Not available

3. Technology Stack

Technology	Purpose
Python	Core development language
Pandas and NumPy	Chunk processing, cleaning, aggregation, and analysis
Scikit-learn	HistGradientBoostingRegressor forecasting model
Joblib	Saved model persistence
Streamlit	Interactive dashboard
Plotly	Interactive demand forecast and risk charts
Matplotlib	EDA charts
GitHub + Streamlit Community Cloud	Source control and public deployment

4. Solution Architecture

The architecture follows a simple, reproducible flow from raw retail transactions to business decisions.

Raw retail data	Clean and validate	Weekly SKU demand	Forecast + backtest	Risk scores + dashboard
-----------------	--------------------	-------------------	---------------------	-------------------------

Pipeline detail: the transaction source is processed in chunks; valid records are grouped by sku_id and weekly date; leakage-safe lag and rolling features are created; time-based validation selects the lower-WAPE method; eight-week forecasts feed the risk engine and Streamlit dashboard.

5. Forecasting Methodology

Weekly Demand Target

The forecasting target is weekly SKU demand: SKU + week -> total units sold. The processed dataset contains sku_id, week, demand, and revenue.

Features and Leakage Control

- Lags: lag_1, lag_2, lag_4, and lag_8.
- Rolling features: shifted four- and eight-week rolling mean, plus shifted four-week rolling standard deviation.
- Calendar features: week number, month, and quarter.
- All rolling statistics use prior demand only, using shift(1) before calculating rolling values.

Validation and Model Selection

A time-based holdout was used rather than random train-test splitting. The final 12 weeks were held out for evaluation. The baseline is an 8-week seasonal-naive lag; the machine-learning alternative is a global HistGradientBoostingRegressor trained across SKUs. WAPE is the primary model-selection metric.

Model	WAPE	MAE	RMSE	Bias
Seasonal naive (8-week lag)	39.86%	8.79	28.44	-7.93%
HistGradientBoostingRegressor	39.25%	8.66	88.83	-1.22%

Selection decision: HistGradientBoostingRegressor was selected because it achieved the lower WAPE (39.25% versus 39.86%) and lower MAE. Its RMSE was higher, so high-demand SKUs should be monitored for larger individual errors.

6. Inventory Risk Scoring

Risk scoring uses the supplied inventory snapshot totals by SKU. Since supplier lead time and on-order quantities were not included in the source, the model uses a transparent operating assumption: a two-week lead time, represented as 25% of the eight-week forecast. Sales at risk and capital locked are estimates, not original source fields.

Decision category	Actual count	Decision rule
REORDER NOW	903	High stockout and low overstock risk
MARKDOWN / CLEAR	2,713	Low stockout and high overstock risk
WATCH / VOLATILE	0	High stockout and high overstock risk
HEALTHY	1,384	Low stockout and low overstock risk

Estimated business impact: sales at risk = 71,091,980.08 monetary units; capital locked = 1,264,675,668.49 monetary units.

7. Dashboard and Visual Evidence

The Streamlit dashboard provides five business-oriented views: Overview, Forecast, Risk, Decision Grid, and Insights. It loads saved output files and the persisted model rather than retraining when the app starts.

Figure 1. FORESIGHT dashboard overview (local Streamlit application).



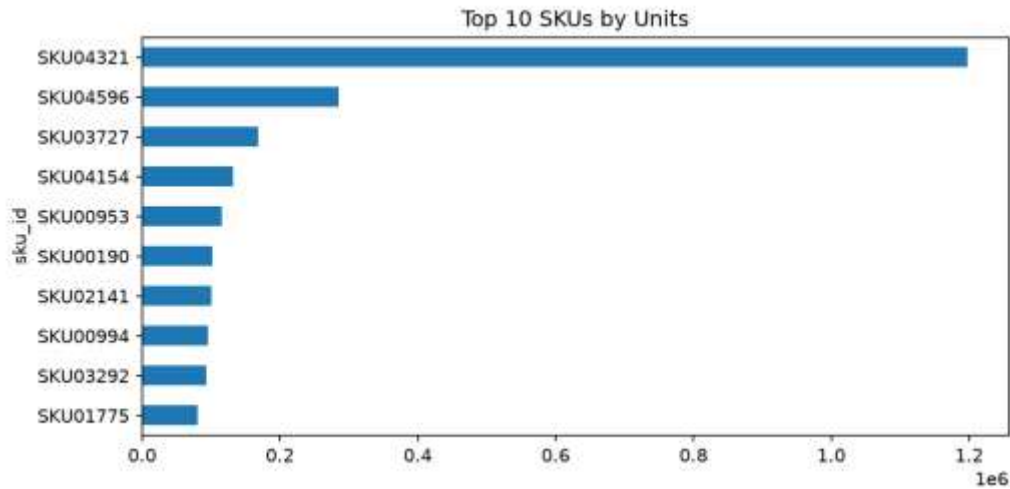


Figure 3. Top 10 SKUs by total units sold.

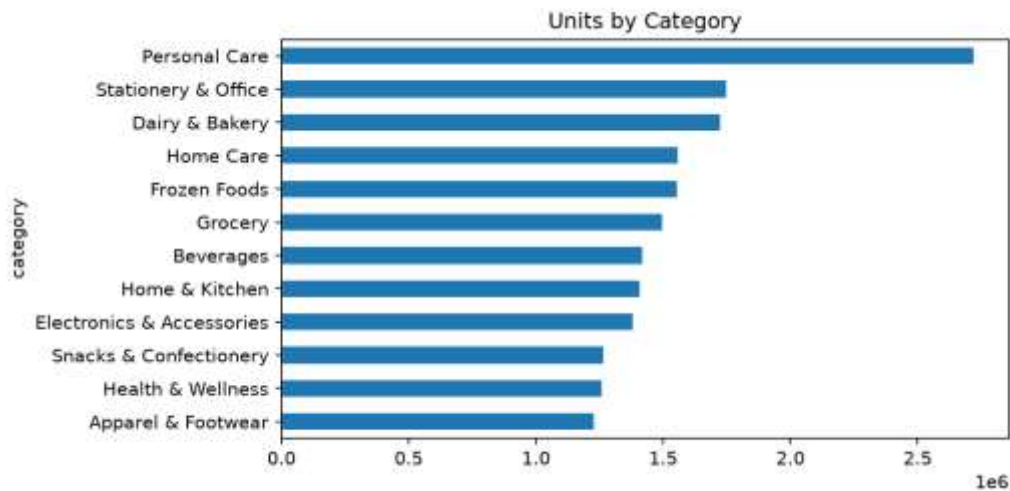


Figure 4. Demand by category.

8. Key Insights and Conclusion

- The source covers nearly 10 million clean retail transactions across four calendar years and 5,000 SKUs.
- SKU04321 is the highest-volume SKU with 1,197,375 units; SKU01863 is the slowest with 1,105 units.
- The model improves primary WAPE modestly over the seasonal-naive baseline, so it is selected for the eight-week outlook while error monitoring remains important.
- The inventory snapshot identifies 903 reorder candidates and 2,713 markdown/clear candidates, supporting focused supply and merchandising review.

9. Conclusion

FORESIGHT delivers a working end-to-end demo: a memory-efficient data pipeline, weekly SKU demand forecasts, honest time-based evaluation, inventory risk scoring based on supplied snapshot data, and an interactive dashboard. The project is transparent about its constraints, including the lack of on-order data, inventory history, and supplier lead-time fields.